Symmetries in Physics — Exercise Sheet 11

$\mathrm{SO}(3)$, the double cover, and spherical harmonics

Variant: Questions only

Exercise 1 Prove the Euler decomposition: every $R \in \mathrm{SO}(3)$ can be written $R = R_z(\alpha)R_y(\beta)R_z(\gamma)$. Identify where the decomposition fails to be unique.

Exercise 2 For $U = \cos\frac{\theta}{2}\,\mathbb{1} - \mathrm{i}\sin\frac{\theta}{2}\,\hat{n}\cdot\vec{\sigma} \in \mathrm{SU}(2)$, compute $\Phi(U)$ acting on $\vec{x}$ via $U(\vec{x}\cdot\vec{\sigma})U^\dagger$, and verify both the Rodrigues formula $\Phi(U)\vec{x} = \cos\theta\,\vec{x} + \sin\theta\,\hat{n}\times\vec{x} + (1-\cos\theta)(\hat{n}\cdot\vec{x})\hat{n}$ (rotation through θ about $\hat{n}$) and the two-to-one property.

Exercise 3 Let P_ℓ be the homogeneous polynomials of degree ℓ on $\mathbb{R}^3$. (i) Show $\dim P_\ell = \binom{\ell+2}{2}$. (ii) Using the pairing of Lecture 11 (the *Fischer pairing*), in which $\langle r^2 f, g\rangle = \langle f, \Delta g\rangle$, prove that $\Delta\colon P_\ell \to P_{\ell-2}$ is surjective, and deduce $\dim H_\ell = 2\ell+1$ for the harmonic subspace.

Exercise 4 Starting from the highest-weight vector $w_2 = (x+\mathrm{i}y)^2$, generate the five harmonics of $\ell = 2$ by repeated application of $L_- = -z(\partial_x - \mathrm{i}\partial_y) + (x-\mathrm{i}y)\partial_z$, and check the chain terminates.

Exercise 5 Prove directly — following the loop $R_z(\theta)$, $\theta \in [0, 2\pi]$ — that a continuous representation of $\mathrm{SO}(3)$ cannot contain half-integer spin.

Exercise 6 A particle on the unit sphere has $H_0 = L^2/2I$. (i) State the spectrum and degeneracies. (ii) A weak field adds $V = \varepsilon\cos^2\theta$, breaking $\mathrm{SO}(3)$ to rotations about $\hat{z}$ (and reflections). Compute the first-order splitting of the $\ell = 1$ level and exhibit the residual degeneracy.

Exercise 7 The spin-j Wigner matrix is $D^j_{m'm}(\alpha,\beta,\gamma) = e^{-\mathrm{i}m'\alpha}\,d^j_{m'm}(\beta)\,e^{-\mathrm{i}m\gamma}$, with the reduced matrix $d^j(\beta) = e^{-\mathrm{i}\beta J_y}$ in the spin-j representation. (i) For $j = 1$, in the basis $m = 1, 0, -1$, compute $d^1(\beta)$ explicitly. (ii) Read off the Euler-angle phase structure of D^1 and verify that D^1 is unitary.